Ten Layers of the Endpoint Spectral-Realizability Frontier

Bin Wang

July 2026

Abstract

RH-312–RH-321 resolve the synthetic endpoint spectral class without promoting it to the actual noisy operator. The deterministic endpoint is a universal logarithm plus an analytic remainder, its best degree-N Hardy error is $N^{-1/2}$, and complete root-of-unity packets realize every finite prefix as a genuine integer-multiplicity normal spectrum. The least rank and squared mass grow like $(q_*/q)^N$, yielding the sharp endpoint law $E_{\mathrm{spec}}(M)^2 \sim \log(q_*/q)/\log M$ and the optimal strict-annulus rates. An escaping packet then proves that fixed moments and every strict annulus can converge while endpoint H^2 fails. Thus actual coefficient transport and endpoint energy tightness remain the precise missing inputs. This batch is therefore a scoped spectral-realizability route stop, not a reopening input. The typed ledgers stay four of five and Gates A–E remain open.

1 Endpoint analytic structure

RH-312 proves

$$-\sum_{n\geq 2} \frac{a_n \rho_*^n}{n} w^n = \log(1-w) + w + H_{\mathrm{reg}}(w),$$

where H_{reg} is analytic past the unit circle. RH-313 splits the logarithm into orthogonal even and odd channels, and RH-314 gives the exact best degree-N error

$$E_N^2 = \sum_{n>N} n^{-2} \sim N^{-1}.$$

These are deterministic Hardy theorems, not noisy convergence.

2 Genuine finite spectral realizability

RH-315 introduces the packet formed from all roots of $z^d = w/(dL)$. It has zero moments below d, moment w at order d, and integer multiplicities. RH-316 recursively realizes every deterministic prefix by a finite conjugate spectrum in $|\mu| \leq q$. RH-317 proves that the optimal rank and squared mass both obey

$$\Theta(s^N), \qquad s = q_*/q = 1.4017504517\ldots.$$

RH-318 converts that complexity clock into the exact endpoint extremal law

$$E_{\mathrm{spec}}(M)^2 \sim \frac{d}{\log M}, \qquad d = \log s = 0.3377217783\ldots.$$

In particular the actual mass cap implies the class-sharp universal lower-rate bound

$$\liminf_{\sigma\downarrow 0} \log(1/\sigma)\|g_\sigma\|^2_{H^2(\rho_*)} \geq d,$$

whose corresponding norm lower scale still tends to zero.

3 Strict annuli versus endpoint energy

RH-319 uses the same finite spectra to attain, for every $1.4 < \rho < \rho_*$,

$$\Theta_\rho\left(M^{-\kappa(\rho)}/\log M\right), \qquad \kappa(\rho) = \frac{\log(1/(q_*\rho))}{\log(q_*/q)}.$$

This upgrades RH-306 from abstract coefficient saturation to genuine normal spectral power sums.

RH-320 adds one escaping packet to an exact prefix. Every fixed moment and every strict-annulus logarithm converge, the inverse-noise mass cap is respected after reparametrization, but one endpoint coefficient stays of order one. Therefore endpoint convergence requires exactly:

$$e_{\sigma,n} \to 0 \text{ for each fixed } n, \qquad \lim_{N\to\infty} \limsup_{\sigma\downarrow 0} \sum_{n>N} |e_{\sigma,n}|^2 = 0.$$

Neither input is proved for the actual complement.

4 Typed frontier

branch	head	bridge	tail	target	boundary	score
noisy modulus spectrum	1	0	1	1	1	4
graded monodromy counterloop	1	1	0	1	1	4

The weighted cross-branch glue remains false and the complete count is zero. Synthetic normal matrices do not fill the missing actual bridge coordinate.

Proposition 4.1 (two remaining inputs)**.** *The direct route reopens only with actual fixed-order complement transport and endpoint energy tightness, equivalently actual endpoint H^2 convergence. The typed route reopens with the joint first-alias boundary-layer trace law identified after RH-311. No finite-prefix or synthetic realization substitutes for either input.*

Proof. The direct statement is the RH-320 ℓ^2 tightness criterion. The typed statement is the RH-310 first-alias obligation, unchanged by the present synthetic constructions. □

Remark 4.2. No Gate-A determinant identification is completed. Gates B–E remain false/open. The batch constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace formula or zeta-divisor equality, and does not imply RH.